

5.07 Non-parametric tests			
5.07a	Non-parametric tests		a) Understand what is meant by a non-parametric hypothesis test, appreciate situations where such tests are useful and be able to select an appropriate test.
5.07b	The basis of non-parametric tests		b) Understand the basis of sign tests, the Wilcoxon signed-rank test and the Wilcoxon rank-sum test (also known as the Mann-Whitney U test). <i>Tables of critical values of T and W will be provided. Learners should know the notation W_+ and W_-.</i>
5.07c	Single-sample hypothesis tests		c) Be able to test a hypothesis concerning a population median using a single-sample sign test and a single-sample Wilcoxon signed-rank test. <i>[Problems in which observations coincide with the hypothetical population median are excluded.]</i>
5.07d 5.07e	Paired-sample and two-sample hypothesis tests		d) Understand the difference between a paired-sample test and a two-sample test, and be able to select the appropriate form when solving problems. e) Be able to test for medians or identity of population as appropriate, using a paired-sample sign test, a Wilcoxon matched-pairs signed-rank test and (for unpaired samples) a Wilcoxon rank-sum test. <i>[Problems involving tied ranks are excluded.]</i>

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
5.07f	Normal approximations		f) Be able to carry out tests using the Wilcoxon signed-rank test and the Wilcoxon rank-sum test for large samples using the approximations: Wilcoxon signed-rank test $T \sim N\left(\frac{1}{4}n(n+1), \frac{1}{24}n(n+1)(2n+1)\right)$ Wilcoxon rank-sum test (samples of sizes m and n, with $m \leq n$) $W \sim N\left(\frac{1}{2}m(m+n+1), \frac{1}{12}mn(m+n+1)\right).$ <i>Includes the use of continuity corrections.</i>